

ALAN MAN

SOFTWARE ENGINEER · FULL STACK DEVELOPER

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience building enterprise web and mobile applications for hospitality and insurance. Experienced with React, TypeScript, Python/FastAPI, Java Spring Boot, SQL, Microsoft Azure and React Native. Build REST APIs, secure workflows, offline-capable applications and cloud deployments. Contributed to a hotel case management platform that helped shift most customer enquiry responses from 1–2 days to the same day. Authorized to work in Canada.

TECHNICAL SKILLS

Languages	TypeScript · JavaScript · Python · Java · SQL
Frontend & Mobile	React · React Native · Redux Toolkit · Material UI · Vite · Angular · HTML · CSS
Backend	FastAPI · Spring Boot · NestJS · Azure Functions · REST APIs
Databases & Data	Azure SQL Database · SQL Server · IBM DB2 · MongoDB · Realm.js · SQL query optimization · SqlPackage CLI
Cloud & DevOps	Azure App Service · Azure Storage Queue · Azure DevOps YAML Pipelines · Azure Key Vault · Managed Identity · Deployment slots · Microsoft Graph API · Jenkins
Security & Testing	Microsoft Entra ID · MSAL · JWT · RBAC · Pydantic · Pytest · Vitest · Integration testing · UAT

PROFESSIONAL EXPERIENCE

May 2025 – Present

System Developer

Eastech Systems Limited · Remote

Core developer of Mercury, an AI-powered enterprise hotel messaging and case management platform for an international hospitality organization.

Key Achievements & Contributions

- **Established an enterprise-scale hotel messaging and case management platform**, processing 62,000+ emails and 20,000+ cases during Q1–Q2 2026.
- **Drove measurable operational improvements**, reducing average email processing time by **74% (10.8 to 2.8 hours)** and case completion time by **72% (89 to 25 hours)** through centralized workflows and AI-assisted automation.
- **Developed full-stack case management workflows** using React, TypeScript, Python, FastAPI, and

Azure SQL, including case assignment, bulk operations, status tracking, and audit history.

- **Integrated AI-assisted workflows** for email analysis, classification, tagging, summarization, urgency assessment, recommended actions, and draft responses.
- **Designed and implemented granular RBAC** with feature-level permissions, user-role assignments, and administrative tools for access management.
- **Implemented secure enterprise authentication and integrations** using Microsoft Entra ID, MSAL, JWT, and Microsoft Graph API.
- **Optimized SQL queries and backend performance**, engineering asynchronous workflows using Azure Functions and Azure Storage Queue.
- **Engineered and managed Azure cloud infrastructure and production deployments** using Azure App Service, Azure DevOps CI/CD, Key Vault, managed identities, and deployment slots.
- **Developed operational dashboards** for monitoring case activity, agent workloads, and workflow performance.

Tech Stack: React, TypeScript, Python, FastAPI, Azure SQL, Microsoft Graph API, Azure Functions, Azure DevOps.

March 2021 – January 2025

Analyst Programmer

Arch-Partnership Limited · Hong Kong

Developed enterprise web and mobile insurance applications for Prudential, supporting high-volume agency and brokerage operations, digital policy submissions, offline workflows, remote signing, and payment processing.

Key Achievements & Contributions

- **Established enterprise insurance application workflows** serving 300+ agents and brokers, processing an estimated 100,000+ insurance applications over multiple years.
- **Accelerated insurance application processing from approximately 1 week to 1–3 days** through workflow improvements, automated validation, electronic signatures, and streamlined submissions.
- **Streamlined end-to-end policy applications** by developing Angular applications integrated with Prudential's internal systems through REST APIs, data mapping, and business rules.
- **Engineered offline-first mobile applications** using React Native and Realm.js, implementing synchronization validation, conflict detection, and data recovery.
- **Enabled secure remote insurance signing** through VoIP-based signing workflows and facial recognition identity verification.
- **Digitized document and payment processing** through multi-party e-signatures, offline PDF generation, payment gateway integration, and transaction verification.
- **Automated insurance calculations and eligibility checks** using Java Spring Boot and NestJS services integrated with external insurer APIs.
- **Delivered secure enterprise applications** using IBM DB2, MongoDB, JWT authentication, Jenkins, and Azure Pipelines.

Tech Stack: Angular, React Native, Realm.js, Java, Spring Boot, NestJS, REST APIs, IBM DB2, MongoDB.

July 2020 – September 2020

Software Engineer (Part-time)

Woow Moment Limited · Hong Kong

Developed cross-platform mobile applications, backend APIs, and location-based services for Ebatte, a

connected power bank platform, and Run2Gather, a running community application.

Key Achievements & Contributions

- **Enabled mobile-to-device connectivity** for Ebatte's power bank platform by developing React Native native-module bridges to integrate JavaScript application logic with native mobile functionality and Bluetooth beacon communication.
- **Engineered RESTful backend APIs** using Node.js and Express.js to facilitate communication between mobile applications, backend services, and connected power bank devices.
- **Improved frontend maintainability and usability** by developing reusable, responsive React.js components for device integration and management workflows.
- **Integrated GPS-based geolocation and Google Maps SDK** into Run2Gather, enabling location-based running activities, route visualization, and interactive mapping.
- **Developed community-focused mobile features** for Run2Gather, supporting running event organization and activity tracking.

Tech Stack: React Native, React.js, JavaScript, Node.js, Express.js, REST APIs, Bluetooth Beacons, Google Maps SDK, GPS, Geolocation Services.

June 2020 – January 2021

Student Helper, Path Advisor

Hong Kong University of Science and Technology (HKUST) · Hong Kong

Contributed to **Path Advisor**, HKUST's campus mapping and indoor navigation platform, integrating location-based services, indoor positioning technologies, and interactive navigation features.

Key Achievements & Contributions

- **Enhanced campus navigation and route-finding functionality** by developing and maintaining React.js features for interactive campus maps, building navigation, and route search.
- **Advanced indoor localization in GPS-denied environments** by contributing to a crowdsourced Indoor Positioning System (IPS) leveraging Wi-Fi Received Signal Strength Indicator (RSSI), Bluetooth Low Energy (BLE) beacon localization, and multi-sensor fusion for indoor positioning and floor-level identification.
- **Developed RESTful backend services** using Node.js, Express.js, and MongoDB to support location data retrieval, route search, and campus mapping, integrating Realm DB for application-side geolocation data.
- **Improved indoor mapping coverage and navigation accuracy** by collecting wireless signal measurements, capturing Ricoh 360° imagery, and maintaining floor plans, building layouts, and pedestrian routes.
- **Improved application reliability and usability** through feature testing, route validation, defect identification, and troubleshooting across frontend and backend components.

Tech Stack: React.js, JavaScript, Node.js, Express.js, MongoDB, Realm DB, Indoor Positioning Systems (IPS), Wi-Fi RSSI, BLE Beacons, Sensor Fusion, Geolocation, REST APIs.

September 2018 – August 2019

IT Student Helper

Audit Commission, HKSAR Government · Hong Kong

Provided IT support and infrastructure administration for the Audit Commission, covering Windows Server environments, identity management, internal web application hosting, and data backup.

Key Achievements & Contributions

- **Maintained reliable daily IT operations** by administering Windows Server environments and troubleshooting workstation, network, email, and user access issues for government staff.
- **Supported secure user access and centralized identity management** through Active Directory administration, account provisioning, and access permission management.
- **Supported internal web application hosting and infrastructure availability** by configuring and maintaining Apache Tomcat, Microsoft IIS, and IBM WebSphere Application Server environments, alongside VMware-based virtual server setup and administration.
- **Strengthened data protection and business continuity** by supporting NAS-based backup solutions and Windows Server replication infrastructure, helping maintain data availability and recovery capabilities for departmental systems.

Tech Stack: Windows Server, Active Directory, Apache Tomcat, Microsoft IIS, IBM WebSphere Application Server, VMware, NAS Backup, Windows Server Replication.

OTHER EXPERIENCE

October 2026

BuildYYC Buildathon

Team Participant · Calgary, AB

- Placed 4th overall at the BuildYYC Buildathon in Calgary with a 5-person team, taking PlateWise from idea to final solution.
- Identified the ideal customer profile (ICP) and ran customer validation with 15–20 potential customers to confirm the problem and demand.
- Assessed the market opportunity and analyzed competitors to define clear points of differentiation.
- Shaped the final solution and established the business model, go-to-market plan and product roadmap.

EDUCATION

Bachelor of Engineering, Computer Engineering — 2021
Hong Kong University of Science and Technology

Associate in Engineering — 2015
Hong Kong Community College